

INFO490 Analytics Dashboard

Memoria | <https://mira-ydqq.onrender.com/chat/analytics/> | test login: mohitg2 / uiuc12345

Snapshot 2026-04-17. 1,641 requests, 70 memory bullets, \$0.5967 spend across 90 days.

Part 1.1 — Analytics questions (10 across 3 categories)

System Performance

- A1. How does end-to-end latency vary across requests over a 30-day window?
- A2. Which model contributes most to overall response time?
- A3. What fraction of requests fail or fall back, and how does that pattern correlate with traffic spikes?

User Behavior

- B1. How is the user's memory store distributed across semantic, episodic, and procedural types?
- B2. How active is each user across days, weeks, and months?
- B3. What is the strength distribution of stored memories, and what fraction is below 50?

Cost

- C1. What is the average cost per request, and how does it move day over day?
- C2. Which model accounts for the largest share of spend, and is that proportional to value?
- C3. What is the prompt-vs-completion token split, and how does that ratio behave over time?
- C4. What is the projected monthly burn at the current trajectory?

Part 1.2 — Widget design plan (question to widget)

Q	Widget	Type
A1	Latency Over Time + Avg/P95/Total/Error KPI cards	Line chart + 4 KPI cards
A2	Latency by Model	Bar chart
A3	Request Status Distribution + Breakdown table	Pie chart + table
B1	Memory Type Distribution + Semantic/Episodic/Procedural KPIs	Pie chart + 3 KPI cards
B2	Conversation Activity + Sessions/Messages KPIs + grouped table	Line chart + KPIs + table
B3	Memory Strength Distribution + Avg Strength KPI	Bar chart + KPI card
C1	Daily Cost Trend + Avg Cost/Request KPI	Area chart + KPI card
C2	Cost by Model + breakdown table	Bar chart + table
C3	Token Usage Breakdown + Prompt/Completion KPI	Stacked bar + KPI
C4	Total Cost + Total Tokens KPIs + 30-day trend	KPI cards + line

Part 1.3 — Statistical thinking

Central tendency and dispersion. Latency cards report mean and p95 together because mean alone hides the long tail; both are computed inside `_get_performance_metrics` by sorting the latency series and indexing the 50th and 95th percentile elements.

Distributions over categorical and numeric axes. Memory Type and Request Status pies are categorical histograms; the Memory Strength bar bins a numeric variable into 25-point ranges. Together they make imbalance visible at a glance.

Trends over time and group comparisons. Latency Over Time, Conversation Activity, and Daily Cost Trend bucket events by day across a 30-day window. Group comparisons in Latency-by-Model, Cost-by-Model, and Token Usage Breakdown answer 'which group is the outlier' without forcing a tab switch.

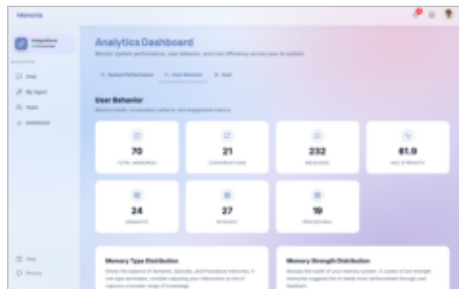
Part 2.4 — Dashboard screenshots (mohitg2, 2026-04-17)

Three tabs of the live dashboard at /chat/analytics/, captured at 1440x900 via Playwright.



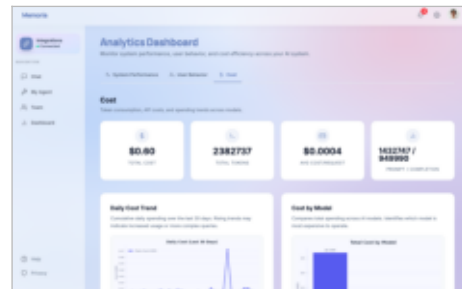
System Performance

docs/screenshots/analytics/01_performance_tab.png



User Behavior

docs/screenshots/analytics/02_behavior_tab.png



Cost

docs/screenshots/analytics/03_cost_tab.png

Submission metadata

Live URL <https://mira-ydqq.onrender.com/chat/analytics/>

Local URL <http://127.0.0.1:8000/chat/analytics/>

Test login username: mohitg2 | password: uiuc12345

Re-seed (local) `python manage.py seed_analytics --user mohitg2 --days 90`

Sample JSON docs/07_api_demos/analytics_api_sample.json

Screenshots docs/screenshots/analytics/{01,02,03}_*.png

Source design doc docs/08_ai_architecture/A10_ANALYTICS_DASHBOARD.md